

Jaypee University of Engineering and Technology, Guna
B.Tech. (Semester II)
Object Oriented Programming Lab (CS202)
Lab Exercise 4

1. Write a C++ program to find the highest occurring digit in prime numbers in a given range.

Given a range L to R, the task is to find the highest occurring digit in prime numbers lie between L and R (both inclusive). If multiple digits have same highest frequency prints the largest of them. If no prime number occurs between L and R, output -1.

Examples:

Input : L = 1 and R = 20.

Output : 1

Prime number between 1 and 20 are 2, 3, 5, 7, 11, 13, 17, and 19.

1 occur maximum i.e. 5 times among 0 to 9.

2. Write four overloaded function to perform addition of different types as follows: First function to add three integers, second to add three floats, third function to add three doubles, and fourth to add three long doubles. Call these functions from main function.
3. Write a function that accepts two arguments: a string name of a movie and an integer running time in minutes. Provide a default value for the minutes so that if you call the function without an integer argument, the minutes default to 90. Write a main() function that proves you can call the function with a string argument alone as well as with a string and an integer.
4. Create a class rectangle with attributes length and width. Provide member functions that calculate the perimeter and area of the rectangle. Provide member functions to get the values from users and display the values of member variables. Write a program to test the class.
5. Create a class named *Shirt* that has the public data members *collarsize* and *sleeveLength*. Create a class named *Pants* that has the public data members *waistSize* and *inSeam*. Write a program that declares one object of each type *Shirt* and *Pants* and assigns values to the objects' data fields. Write two overloaded functions named *displayClothingFacts()*. One version of the function takes a *Shirt* object as an argument; the other version takes a *Pants* object. Each version displays the facts about the piece of clothing. Your main() function should demonstrate that you can call *displayClothingFacts()* with either type of clothing.
6. Define a class named *Movie*. Include private fields for the title, year, and name of the director. Include three public functions with the prototypes `void Movie::setTitle(string);` , `void Movie::setYear(int);` and `void Movie::setDirector(string);`. Include another function that displays all the information about a *Movie*. Write a main() function that declares a movie object named *myFavoriteMovie*. Set and display the object's fields.

7. Write a class definition for an **order** class for a nightclub that contains a table number, a server's name, and the number of patrons at the table. Include a private static data member for the table minimum charge, which is \$4.75. Write a main() function that declares no object of **order** class type, but uses a static member function to display the table minimum charge.
8. Solve the following questions from Hackerrank:
 - a. <https://www.hackerrank.com/challenges/c-tutorial-struct/problem?isFullScreen=true>
 - b. <https://www.hackerrank.com/challenges/c-tutorial-class/problem?isFullScreen=true>
 - c. <https://www.hackerrank.com/challenges/classes-objects/problem?isFullScreen=true>